

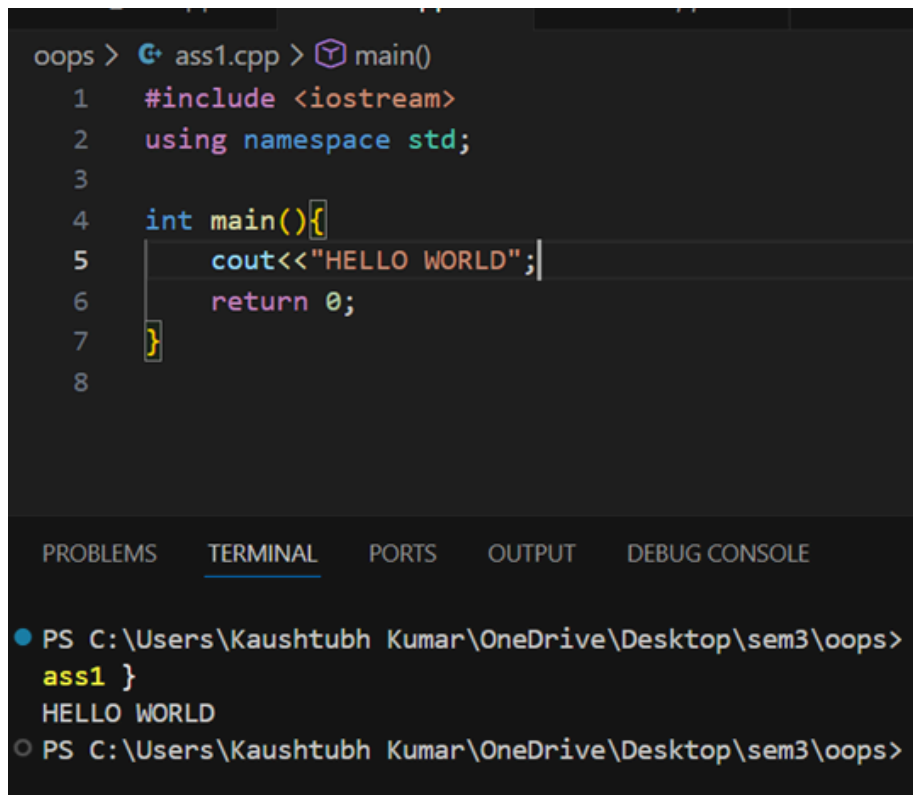
ASSIGNMENT-01

Q1 LAB -1

Topic Covered - cin/cout basics, structures in C, structures in C++, Classes and objects

Programming Question

1. Write a program (WAP) to display "Hello World" on console display.

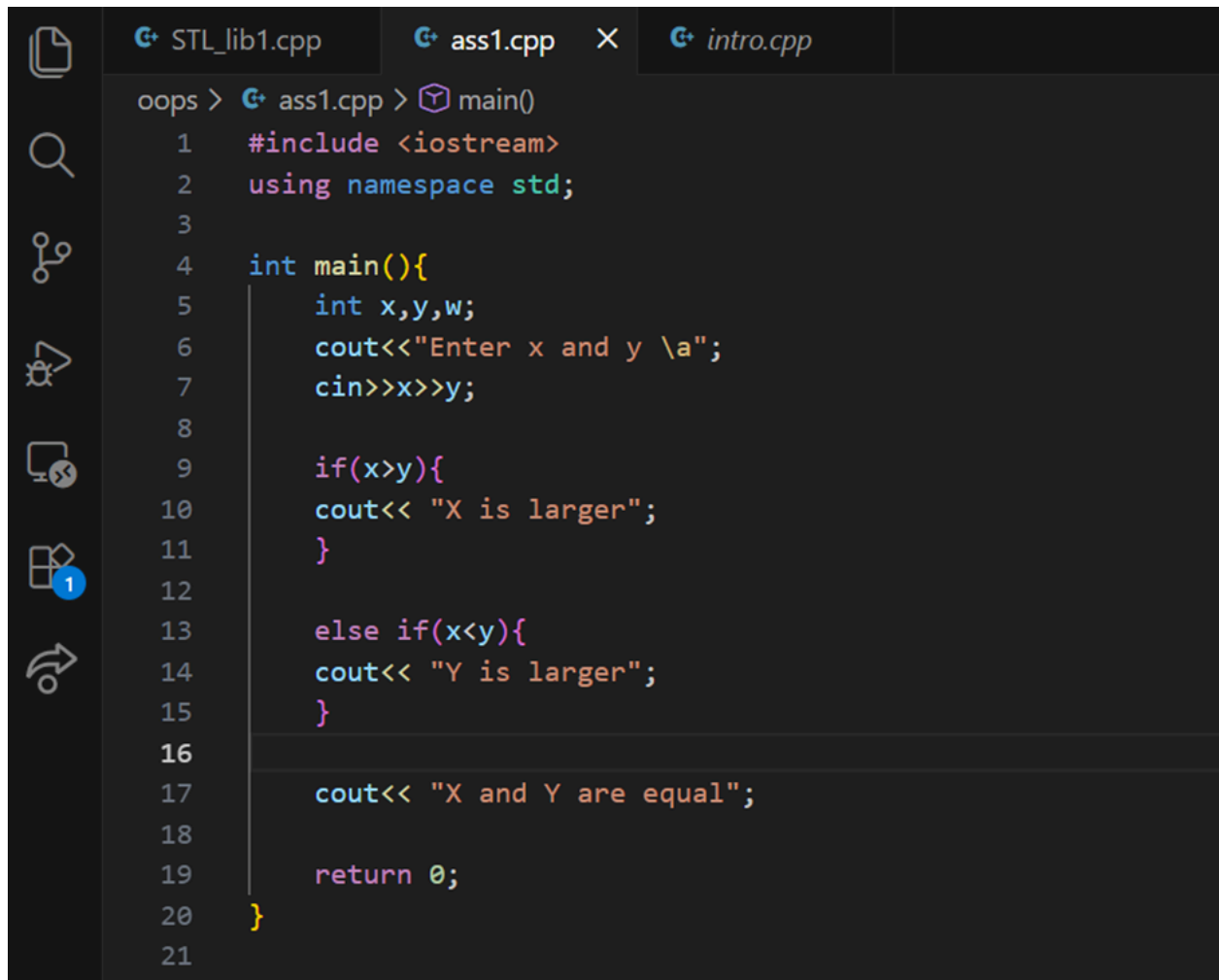


```
oops > G+ ass1.cpp > main()
1  #include <iostream>
2  using namespace std;
3
4  int main(){
5      cout<<"HELLO WORLD";
6      return 0;
7  }
8

PROBLEMS  TERMINAL  PORTS  OUTPUT  DEBUG CONSOLE

● PS C:\Users\Kaushtubh Kumar\OneDrive\Desktop\sem3\oops> g++ ass1.cpp
ass1 }
HELLO WORLD
○ PS C:\Users\Kaushtubh Kumar\OneDrive\Desktop\sem3\oops>
```

2. WAP to that reads two values from keyboard and displays the larger value on the screen.



```
oops > STL_lib1.cpp  ass1.cpp  X  intro.cpp
oops > ass1.cpp > main()
1  #include <iostream>
2  using namespace std;
3
4  int main(){
5      int x,y,w;
6      cout<<"Enter x and y \a";
7      cin>>x>>y;
8
9      if(x>y){
10         cout<< "X is larger";
11     }
12
13     else if(x<y){
14         cout<< "Y is larger";
15     }
16
17     cout<< "X and Y are equal";
18
19     return 0;
20 }
21
```

OUTPUTS:

```
Enter x and y 30
23
X is larger
```

```
Enter x and y 23
30
Y is larger
```

```
Enter x and y 23
23
X and Y are equal
```

3. WAP to read three values a, b, c from keyboard and display the value of $x = a/b - c$.

```
#include <iostream>
using namespace std;

int main() {
    float a, b, c, x;

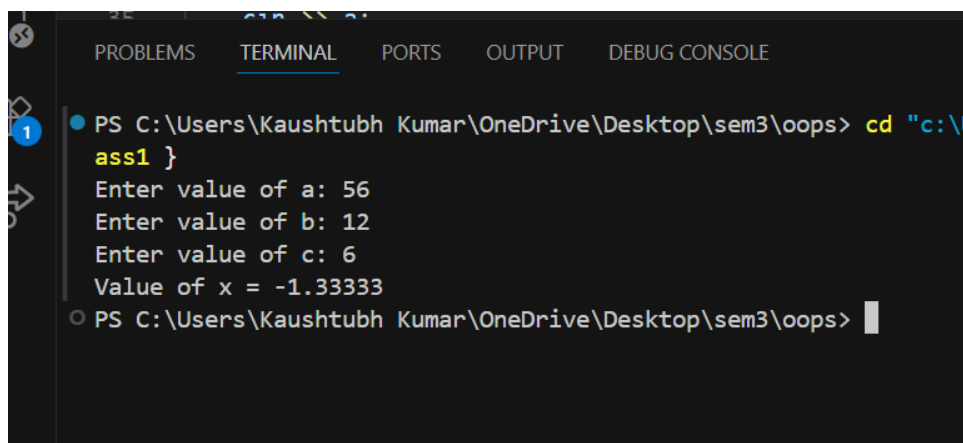
    cout << "Enter value of a: ";
    cin >> a;
    cout << "Enter value of b: ";
    cin >> b;
    cout << "Enter value of c: ";
    cin >> c;

    x = a / b - c;

    cout << "Value of x = " << x << endl;

    return 0;
}
```

OUTPUT:

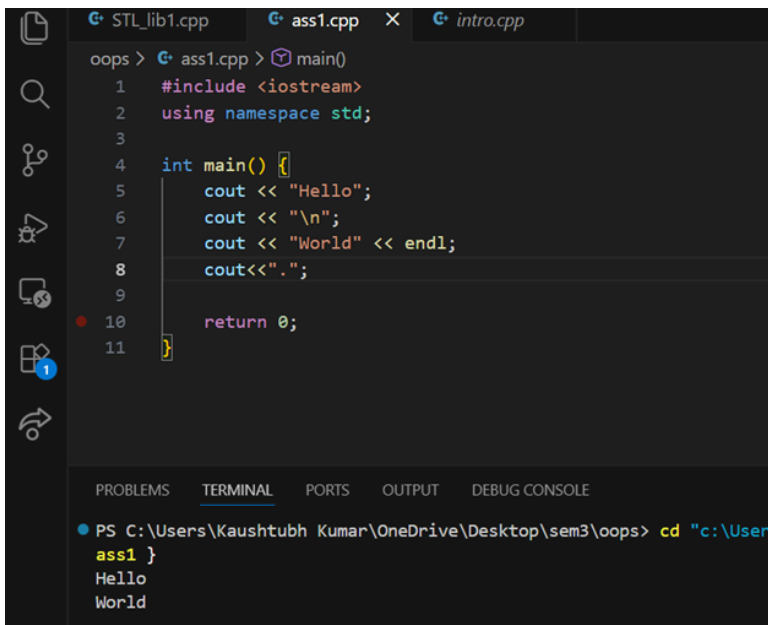


The screenshot shows a terminal window with the following output:

```
PS C:\Users\Kaushtubh Kumar\OneDrive\Desktop\sem3\oops> cd "c:\ass1"
Enter value of a: 56
Enter value of b: 12
Enter value of c: 6
Value of x = -1.33333
PS C:\Users\Kaushtubh Kumar\OneDrive\Desktop\sem3\oops>
```

4. WAP to implement the following control characters:

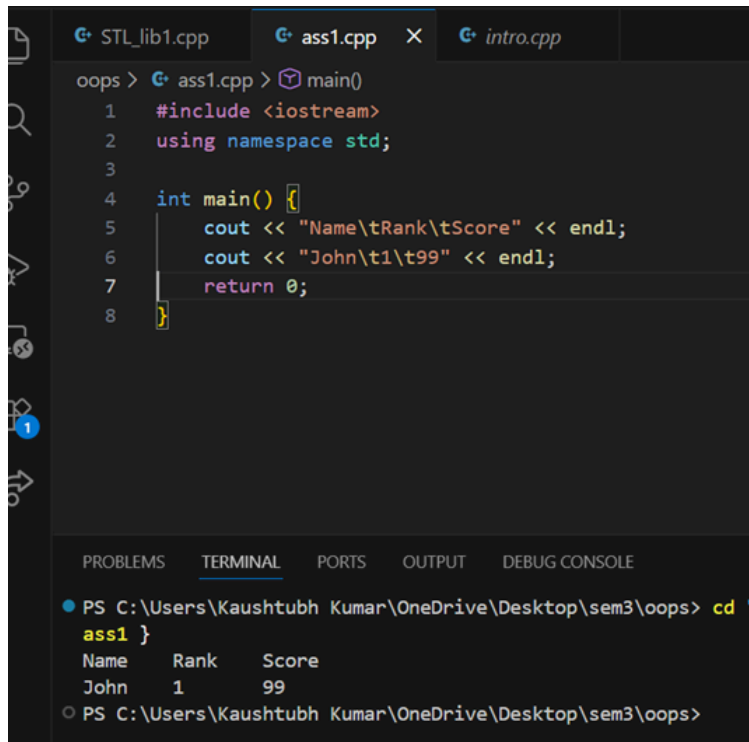
- *'\n' is for new line, or you can use endl –*
cout<<endl<<"message";
- *'\t' is for tab*
- *'\a' is an alarm sound*
- *'\r' is carriage return to go to the beginning of the current line*



```
oops > G+ ass1.cpp > main()
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      cout << "Hello";
6      cout << "\n";
7      cout << "World" << endl;
8      cout<< ".";
9
10     return 0;
11 }
```

PROBLEMS TERMINAL PORTS OUTPUT DEBUG CONSOLE

```
PS C:\Users\Kaushtubh Kumar\OneDrive\Desktop\sem3\oops> cd "c:\User
ass1 }
Hello
World
```

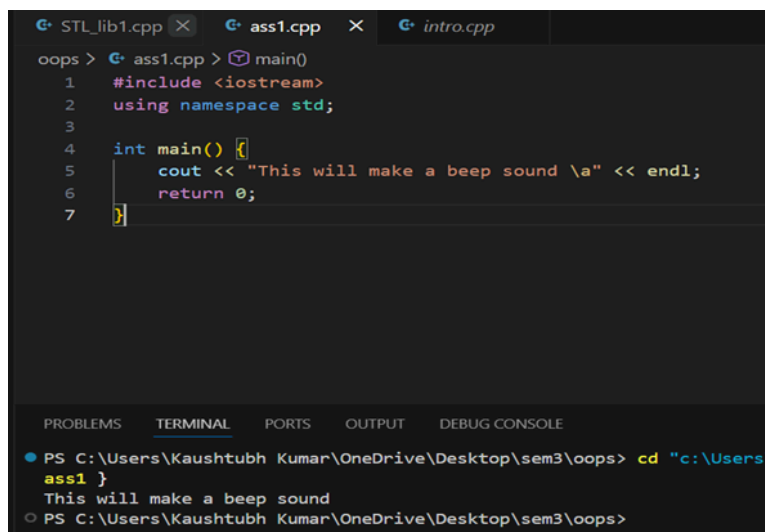


The screenshot shows a C++ IDE with three tabs: STL_lib1.cpp, ass1.cpp, and intro.cpp. The active tab is ass1.cpp, which contains the following code:

```
oops > ass1.cpp > main()
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      cout << "Name\tRank\tScore" << endl;
6      cout << "John\t1\t99" << endl;
7      return 0;
8  }
```

The bottom panel shows the terminal output:

```
PS C:\Users\Kaushtubh Kumar\OneDrive\Desktop\sem3\oops> cd "ass1"
Name    Rank    Score
John    1       99
PS C:\Users\Kaushtubh Kumar\OneDrive\Desktop\sem3\oops>
```

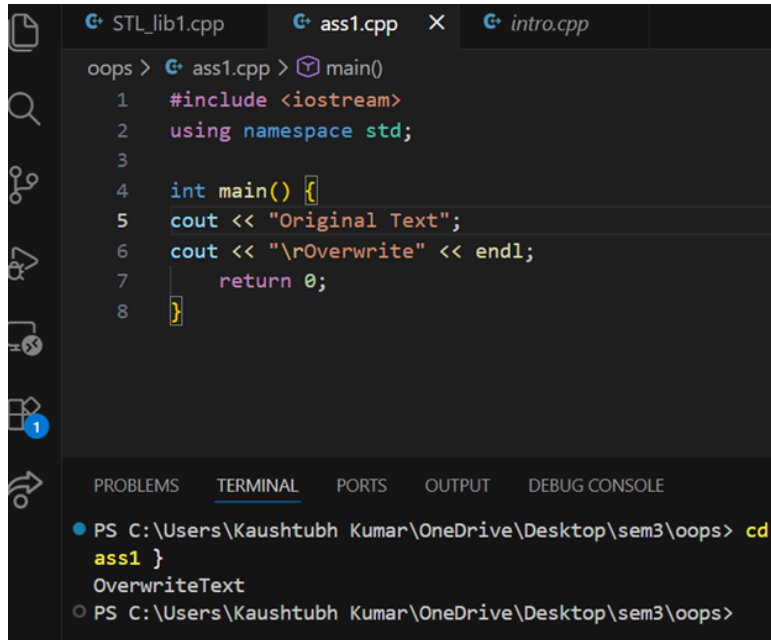


The screenshot shows the same C++ IDE with the same three tabs. The active tab is ass1.cpp, which contains the following code:

```
oops > ass1.cpp > main()
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      cout << "This will make a beep sound \a" << endl;
6      return 0;
7  }
```

The bottom panel shows the terminal output:

```
PS C:\Users\Kaushtubh Kumar\OneDrive\Desktop\sem3\oops> cd "c:\Users\Kaushtubh Kumar\OneDrive\Desktop\sem3\oops"
This will make a beep sound
PS C:\Users\Kaushtubh Kumar\OneDrive\Desktop\sem3\oops>
```

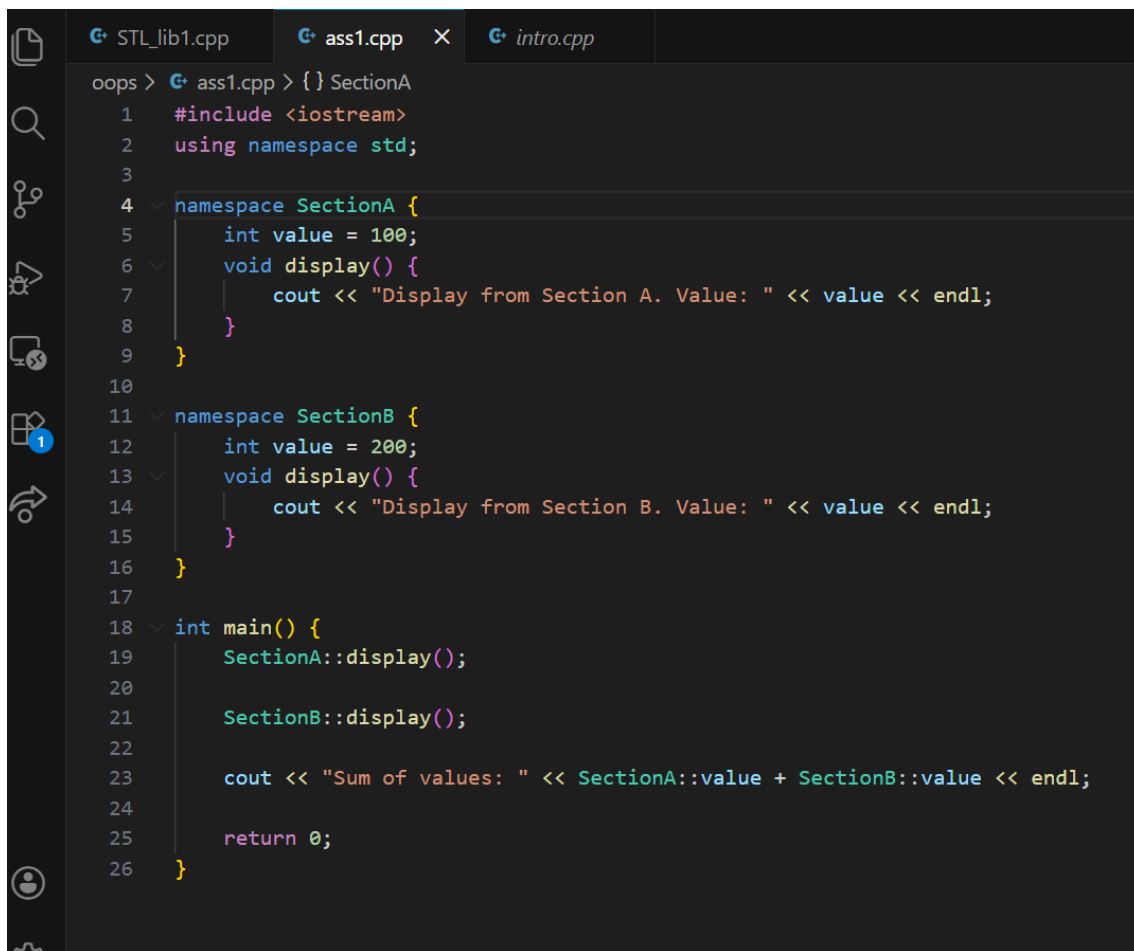


```
oops > STL_lib1.cpp  ass1.cpp  X  intro.cpp
oops > ass1.cpp > main()
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5  cout << "Original Text";
6  cout << "\rOverwrite" << endl;
7      return 0;
8  }
```

PROBLEMS TERMINAL PORTS OUTPUT DEBUG CONSOLE

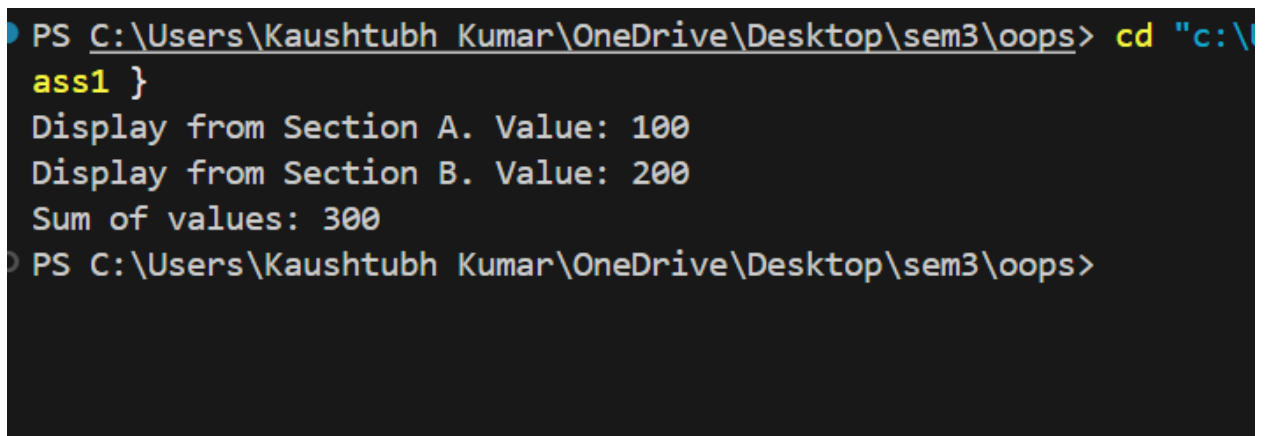
```
● PS C:\Users\Kaushtubh Kumar\OneDrive\Desktop\sem3\oops> cd
ass1 }
OverwriteText
○ PS C:\Users\Kaushtubh Kumar\OneDrive\Desktop\sem3\oops>
```

5. Implement *namespace* in a program to illustrate the use of same name variables and functions in different sections/libraries of the code.

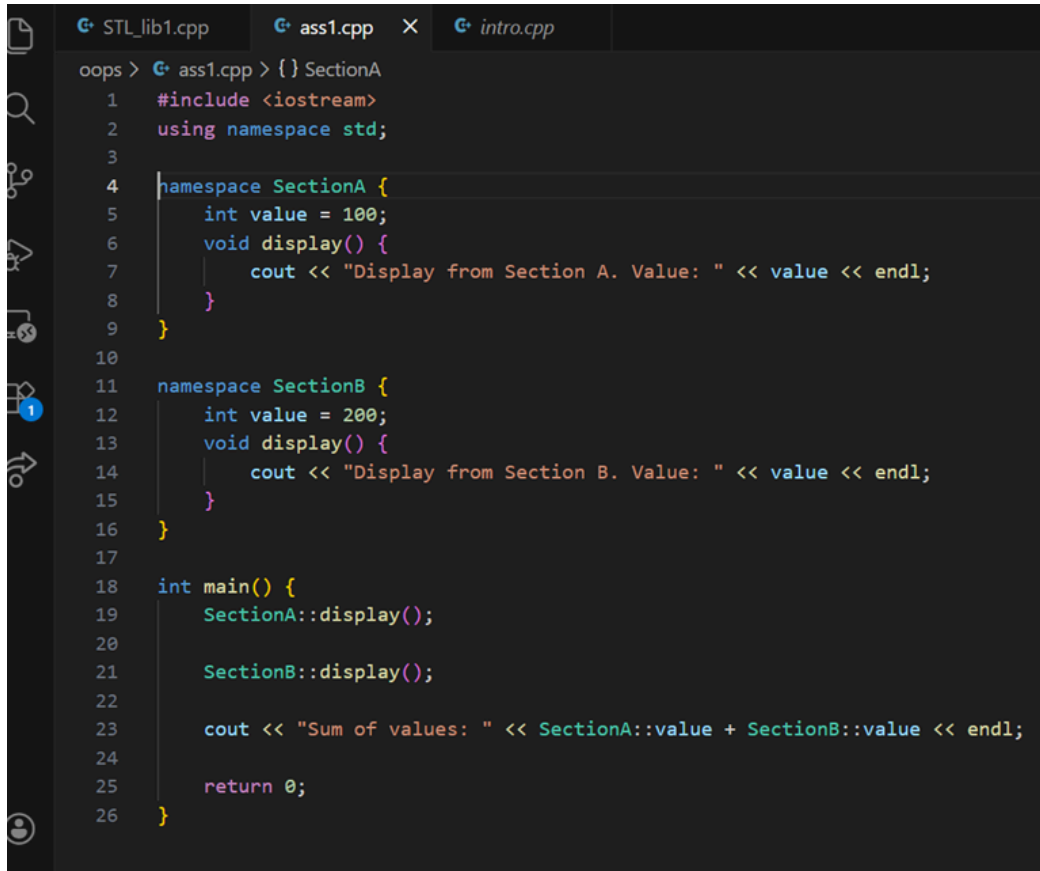


```
oops > STL_lib1.cpp  ass1.cpp x  intro.cpp
oops > ass1.cpp > {} SectionA
1  #include <iostream>
2  using namespace std;
3
4  namespace SectionA {
5      int value = 100;
6      void display() {
7          cout << "Display from Section A. Value: " << value << endl;
8      }
9  }
10
11 namespace SectionB {
12     int value = 200;
13     void display() {
14         cout << "Display from Section B. Value: " << value << endl;
15     }
16 }
17
18 int main() {
19     SectionA::display();
20
21     SectionB::display();
22
23     cout << "Sum of values: " << SectionA::value + SectionB::value << endl;
24
25     return 0;
26 }
```

OUTPUT:



```
PS C:\Users\Kaushtubh Kumar\OneDrive\Desktop\sem3\oops> cd "c:\t
ass1 }
Display from Section A. Value: 100
Display from Section B. Value: 200
Sum of values: 300
PS C:\Users\Kaushtubh Kumar\OneDrive\Desktop\sem3\oops>
```



```
oops > STL_lib1.cpp  ass1.cpp X  intro.cpp
oops > ass1.cpp > {} SectionA
1  #include <iostream>
2  using namespace std;
3
4  namespace SectionA {
5      int value = 100;
6      void display() {
7          cout << "Display from Section A. Value: " << value << endl;
8      }
9  }
10
11 namespace SectionB {
12     int value = 200;
13     void display() {
14         cout << "Display from Section B. Value: " << value << endl;
15     }
16 }
17
18 int main() {
19     SectionA::display();
20
21     SectionB::display();
22
23     cout << "Sum of values: " << SectionA::value + SectionB::value << endl;
24
25     return 0;
26 }
```

6. Write a program to define a structure *Student* that contains 3 variables and 2 functions:

- *Variables:*

(a) *name*

(b) *roll*

(c) *marks*

- *Functions:*

(1) *setStudentData()*

(2) *getStudentData()*

Use loops to input and output data for 3 student object entries.


```
STL_lib1.cpp  ass1.cpp  intro.cpp
oops > ass1.cpp > Student
1  #include <iostream>
2  using namespace std;
3
4  struct Student {
5      string name;
6      int roll;
7      float marks;
8
9      void setStudentData() {
10         cout << "Enter Name: ";
11         cin >> name;
12         cout << "Enter Roll No: ";
13         cin >> roll;
14         cout << "Enter Marks: ";
15         cin >> marks;
16     }
17
18     void getStudentData() {
19         cout << "Name: " << name << " | Roll: " << roll << " | Marks: " << marks << endl;
20     }
21 };
22
```

```
int main() {
    Student students[3];

    cout << "--- Enter Data for 3 Students ---" << endl;
    for(int i = 0; i < 3; i++) {
        cout << "\nStudent " << i+1 << ":" << endl;
        students[i].setStudentData();
    }

    cout << "\n--- Displaying Data ---" << endl;
    for(int i = 0; i < 3; i++) {
        students[i].getStudentData();
    }

    return 0;
}
```

OUTPUT:

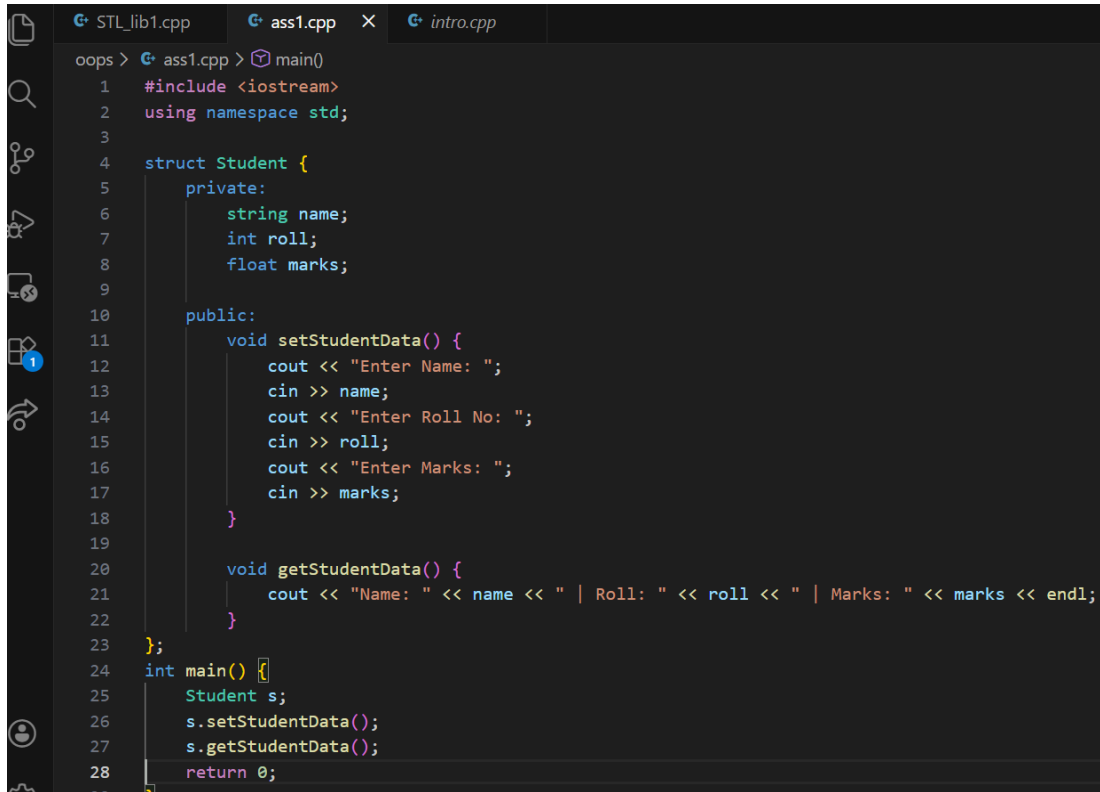
```
Student 1:
Enter Name: ram
Enter Roll No: 12
Enter Marks: 56

Student 2:
Enter Name: tanvi
Enter Roll No: 13
Enter Marks: 50

Student 3:
Enter Name: tushar
Enter Roll No: 14
Enter Marks: 69

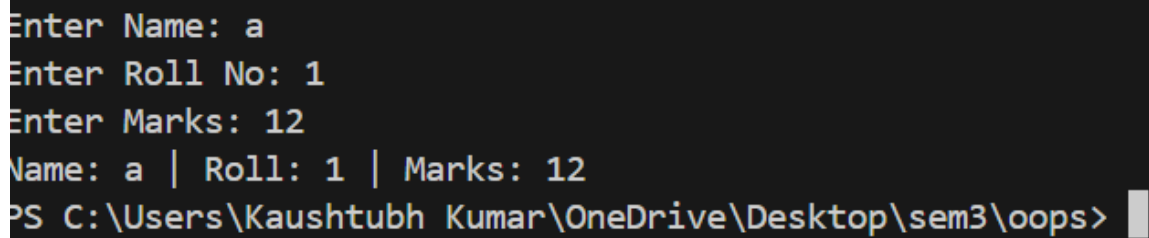
--- Displaying Data ---
Name: ram | Roll: 12 | Marks: 56
Name: tanvi | Roll: 13 | Marks: 50
Name: tushar | Roll: 14 | Marks: 69
PS C:\Users\Kaushtubh Kumar\OneDrive\Desktop\sem3\oops> |
```

7. In the previous program make the data private, and functions public.



```
oops > STL_lib1.cpp  ass1.cpp  intro.cpp
oops > ass1.cpp > main()
1  #include <iostream>
2  using namespace std;
3
4  struct Student {
5      private:
6          string name;
7          int roll;
8          float marks;
9
10     public:
11         void setStudentData() {
12             cout << "Enter Name: ";
13             cin >> name;
14             cout << "Enter Roll No: ";
15             cin >> roll;
16             cout << "Enter Marks: ";
17             cin >> marks;
18         }
19
20         void getStudentData() {
21             cout << "Name: " << name << " | Roll: " << roll << " | Marks: " << marks << endl;
22         }
23     };
24 int main() {
25     Student s;
26     s.setStudentData();
27     s.getStudentData();
28     return 0;
29 }
```

OUTPUT:



```
Enter Name: a
Enter Roll No: 1
Enter Marks: 12
Name: a | Roll: 1 | Marks: 12
PS C:\Users\Kaushtubh Kumar\OneDrive\Desktop\sem3\oops>
```

8. Convert program #4 from struct to class.

```
oops > STL_lib1.cpp  ass1.cpp  intro.cpp
oops > ass1.cpp > main()
1  #include <iostream>
2  using namespace std;
3
4  class Student {
5  string name;
6  int roll;
7  float marks;
8
9  public:
10     void setStudentData() {
11         cout << "Enter Name: ";
12         cin >> name;
13         cout << "Enter Roll No: ";
14         cin >> roll;
15         cout << "Enter Marks: ";
16         cin >> marks;
17     }
18
19     void getStudentData() {
20         cout << "Name: " << name << " | Roll: " << roll << " | Marks: " << marks << endl;
21     }
22 };
23
24 int main() {
25     Student students[3];
26     cout << "Enter Data for 3 Students (Using Class)" << endl;
27     for(int i=0;i<3;i++) {
28         cout << "\nStudent " << i+1 << ":" << endl;
29         students[i].setStudentData();
30     }
31
32     cout << "\n Displaying Data" << endl;
33     for(int i = 0; i < 3; i++) {
34         students[i].getStudentData();
35     }
36     return 0;
37 }
```

OUTPUT:

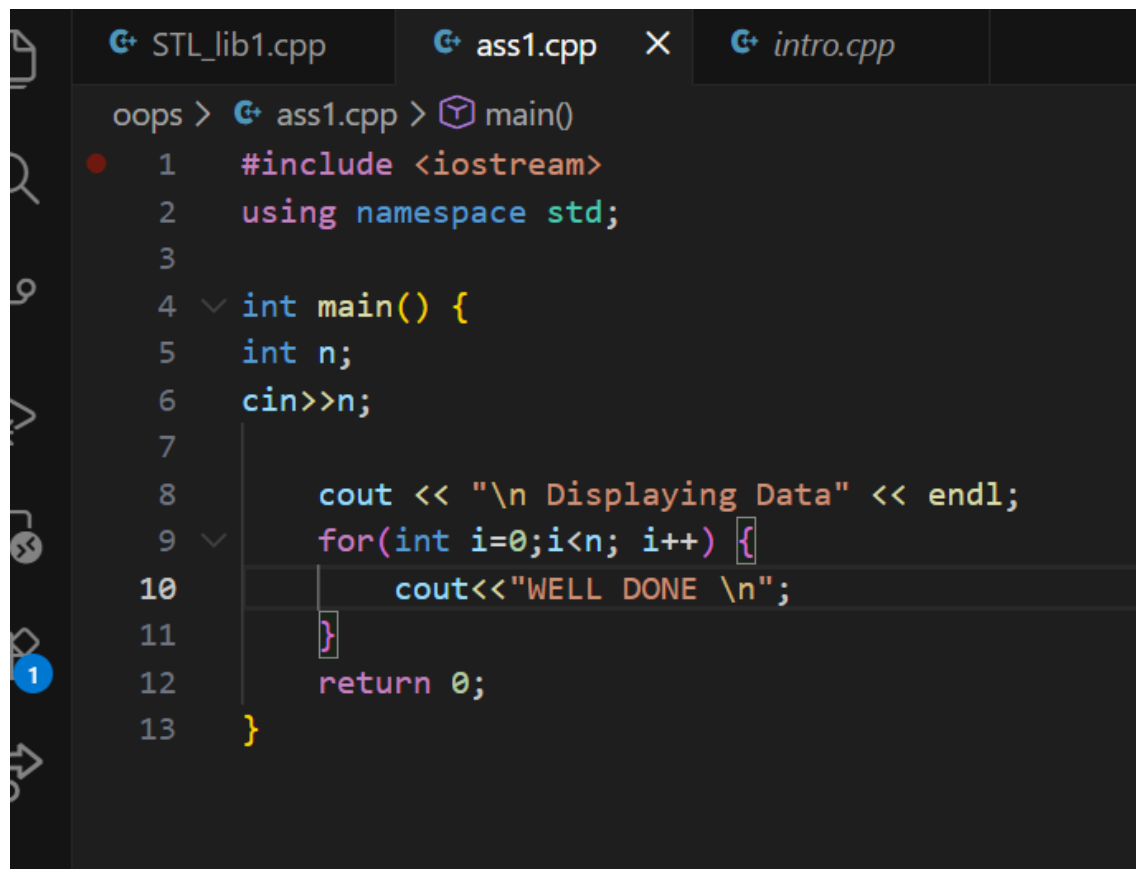
```
Student 1:
Enter Name: aashu
Enter Roll No: 1
Enter Marks: 0

Student 2:
Enter Name: rashu
Enter Roll No: 2
Enter Marks: 12

Student 3:
Enter Name: rishu
Enter Roll No: 3
Enter Marks: 34

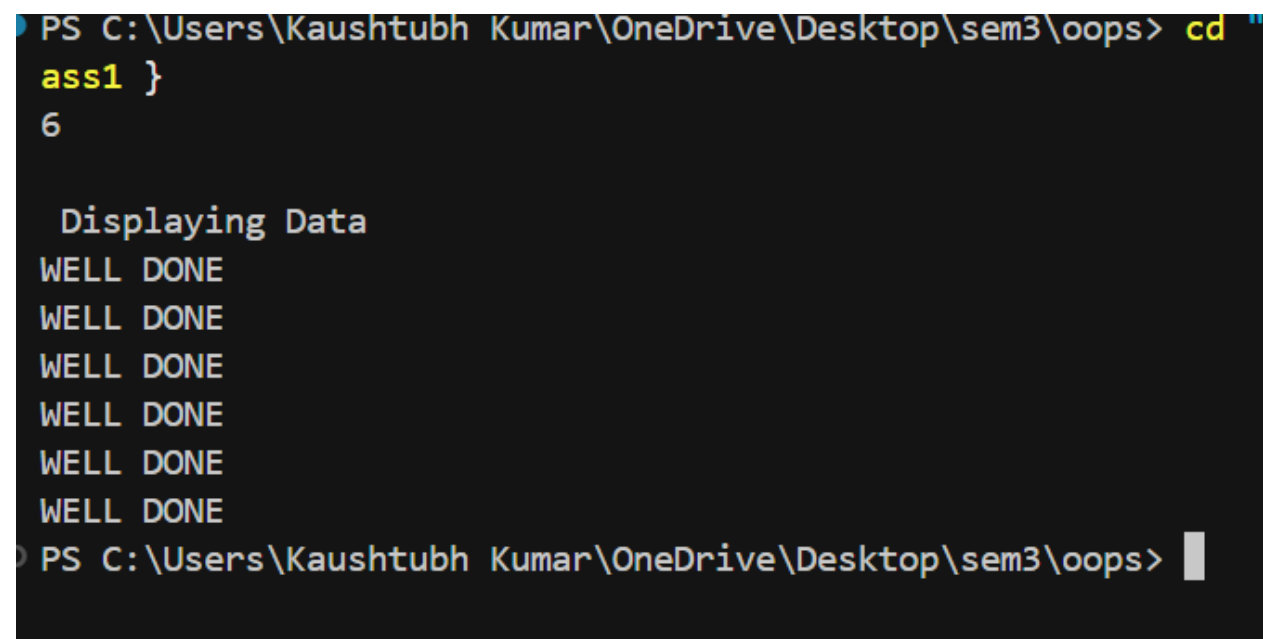
Displaying Data
Name: aashu | Roll: 1 | Marks: 0
Name: rashu | Roll: 2 | Marks: 12
Name: rishu | Roll: 3 | Marks: 34
PS C:\Users\Kaushtubh Kumar\OneDrive\Desktop\sem3\oops>
```

10. WAP to read an integer value from the keyboard and display on screen "WELL DONE" that many times.



```
oops > G+ ass1.cpp > main()
1  #include <iostream>
2  using namespace std;
3
4  int main() {
5      int n;
6      cin>>n;
7
8      cout << "\n Displaying Data" << endl;
9      for(int i=0;i<n; i++) {
10         cout<<"WELL DONE \n";
11     }
12     return 0;
13 }
```

OUTPUT:



```
PS C:\Users\Kaushtubh Kumar\OneDrive\Desktop\sem3\oops> cd "
ass1 }
6

Displaying Data
WELL DONE
WELL DONE
WELL DONE
WELL DONE
WELL DONE
WELL DONE
PS C:\Users\Kaushtubh Kumar\OneDrive\Desktop\sem3\oops>
```

